

Anray Liu

anrayliu@gmail.com | (437) 662-8998 | <https://linkedin.com/in/anray-liu> | <https://github.com/anrayliu>

Profile

Student cloud engineer with 16 months of industry experience in cloud platforms, DevOps, and microservice development. Experienced in building CI/CD pipelines, AWS/Azure infrastructure, and Kubernetes workloads. Proven ability to increase system performance and reduce cloud costs in production. **Seeking Summer 2027 internship.**

Education

University of Ottawa

B.Sc. Honours Computer Science with Co-op (9.5/10.0 GPA)

Ottawa, ON

Fall 2023 – Winter 2028

Work Experience

DevOps Engineer Intern

January 2026 – August 2026

Magnet Forensics

Ottawa, ON

- Conducted performance deep dives into C#/ASP.NET code, DynamoDB, and Lambda functions; identified and resolved bottlenecks, leading to 50% less CPU usage and doubled production server throughput.
- Designed and led the end-to-end implementation of a reverse proxy using AWS CloudFront, eliminating a top-priority sales blocker for clients requiring static IP allowlisting.
- Resolved recurring production outages by analyzing CloudWatch metrics, tuning ECS scaling policies, and implementing algorithmic changes to reduce recovery load (exponential backoff and random jitter).
- Executed a company-wide domain migration across multiple AWS Route 53 hosted zones; maintained clear communication with 3 engineering teams to minimize risk and downtime.
- Deployed a new development environment from scratch using AWS CloudFormation, Route 53, Certificate Manager, and Secrets Manager, while serving as the primary maintainer across all environments.
- Automated load testing using EC2 and Jenkins to validate system scalability under increased customer demand.
- Accelerated feature delivery through a spec-driven workflow using AI assistants (Claude Code, GitHub Copilot) while safeguarding confidential data to ensure security compliance.

Cloud Engineer Intern

May 2025 – December 2025

Statistics Canada

Ottawa, ON

- Developed an open-source Kubernetes microservice using Go (Golang) to automate cleanup of unused Persistent Volumes and Persistent Volume Claims, reducing annual cloud expenses by \$20K.
- Built CI/CD pipelines using GitHub Actions to automate testing, linting, security scans, and Docker containerization.
- Delivered zero-downtime deployments with Argo CD to an Azure Kubernetes Service (AKS) platform of 300+ users.
- Analyzed data processing Python packages using memory profilers, identifying peak usage improvements of 5% across 100GB+ datasets.

Projects

Pyvidplayer2 API | [Link](#)

July 2023 – Present

- Pioneered an open-source Python video library with 60K+ downloads and 3+ years of active maintenance.
- Collaborated with open-source developers to resolve 50+ GitHub issues; developed templates and detailed documentation to maintain industry standards and code quality.
- Automated builds and PyPI releases with GitHub Actions using protected environments and OIDC authentication.

Skills and Certifications

- Skills:** Python, Go, AWS, Azure, Linux, Kubernetes, Docker, Argo CD, Jenkins, GitHub Actions, Azure Pipelines
- Canadian Computing Contest (Senior) – Certificate of Distinction (Top 25%), 2023